Do Anomalies Die When They Are Published? Out-of-Sample and Post-Publication Decay in 212 Cross-Sectional Predictors

Christopher Abhi Dadoo Lei Gao*

July 2026

Abstract

McLean and Pontiff (2016) show that published stock-return anomalies weaken after the academic articles that document them appear. We revisit their question with the full Open Source Asset Pricing library—monthly long-short returns for **212** published predictors, **173,302** anomaly-months over 1926–2024—and classify each anomaly-month as in-sample, post-sample (out of sample but before publication), or post-publication using each study's original sample-end year and journal-publication year. The average anomaly earns **0.61%** per month in sample; this falls to 0.42% out of sample and to **0.30%** after publication—a within-anomaly decline of **−51%** (paired $t = -7.6$). In a pooled panel with anomaly fixed effects the post-publication

*Dadoo: ASSIP 2026, George Mason University. Gao (corresponding author): Costello College of Business, George Mason University, `lgao9@gmu.edu`. This is a student–faculty working paper prepared in the ASSIP 2026 Empirical Finance research group. All returns and metadata are from the public Open Source Asset Pricing dataset (Chen and Zimmermann, 2022); every number in the paper traces to a CSV in `output/tables/` and is reproduced by the code in `code/`. No number is hand-entered or simulated. We thank the ASSIP 2026 cohort for comments. Any errors are our own.

return is 0.27 points per month lower ($t = -7.9$), and the effect survives anomaly-plus-calendar-month fixed effects (-0.24, $t = -3.2$). The decay is thus large and robust. Four extensions sharpen the interpretation without overclaiming. First, the decay is strongly *heterogeneous* in observable strength: anomalies with above-median in-sample returns lose an extra -0.28 points ($t = -4.3$) beyond the baseline, exactly the cross-sectional pattern McLean and Pontiff (2016) predict (though partly regression to the mean). Second, because every anomaly is eventually "treated," we build a *not-yet-published-control* stacked difference-in-differences: it gives a clear -0.37 ($t = -5.2$) with cohort-anomaly fixed effects but only -0.05 ($t = -0.6$) once cohort$\times$calendar-year effects absorb the common trend—so the drop is largely an out-of-sample phenomenon, not a discrete publication kink. Third, *randomization inference* that permutes each anomaly's publication date within its own out-of-sample life shows the true date is not special for the level decay (RI $p = 0.31$) and only marginally so for the incremental effect ($p = 0.08$). Fourth, a *minimum-detectable-effect* analysis shows the null on the risk-versus-mispricing split is informative only for large differences ($\geq$39% of the in-sample mean), which is exactly why the pivotal hand-collection task—reading each original article's stated economic framing—is left to the student.

JEL: G12, G14, G11. **Keywords:** anomalies, market efficiency, return predictability, publication, limits to arbitrage, randomization inference, stacked difference-in-differences.

1 Introduction

A central tension in empirical asset pricing is whether documented cross-sectional return predictors reflect *risk* or *mispricing*. McLean and Pontiff (2016) sharpen the question by asking what happens to an anomaly's long-short return *after* the academic paper documenting it is published. If a predictor is compensation for risk, publicizing it should not change its return; if it is mispricing, sophisticated investors who read the paper should trade against it and shrink it. They find that anomaly returns are 26% lower out of sample and 58% lower post-publication, evidence that academic research itself erodes predictability. This paper re-examines that finding on the largest openly available anomaly library and asks two questions: (i) how large is the out-of-sample and post-publication decay when measured on 212 predictors through 2024, and (ii) is the decay concentrated in mispricing-framed anomalies, as the arbitrage story predicts?

We use the Open Source Asset Pricing dataset of Chen and Zimmermann (2022), which provides monthly long-short returns for 212 published predictors together with a metadata file recording each predictor's original authors, journal, publication year, and sample window. We form a panel of 173,302 anomaly-months (1926–2024) and label each observation *in-sample* (calendar year $\leq$ the study's sample-end year), *post-sample* (after the sample ends but before publication), or *post-publication* (publication year and after), exactly the three regimes of McLean and Pontiff (2016).

Our first result confirms and updates theirs. The equal-weighted average anomaly earns **0.61%** per month in sample, **0.42%** post-sample, and **0.30%** post-publication: a -32% out-of-sample decline and a -51% post-publication decline (paired within-anomaly $t = -7.6$; Table 2). In a pooled panel the post-publication indicator carries a coefficient of -0.24 to -0.27 points per month with t-statistics from -6.4 (no fixed effects) to -7.9 (anomaly fixed effects), and it remains -0.24 ($t = -3.2$) with both anomaly and calendar-month fixed effects (Table 3). Figure 1 plots the level in event-time: it sits near 100% of the in-sample mean until the publication year and then falls to roughly 55–60%, where it stays.

Our second and third results are honest qualifications that a naive reading would miss. *First*, the decline is best described as an *out-of-sample* effect rather than a sharp publication event. Following McLean and Pontiff (2016), we decompose the decline into an "out-of-sample" component (present as soon as the original sample ends) and an "extra-at-publication" component. With anomaly fixed effects the extra publication effect is economically and statistically meaningful (-0.13, $t = -2.2$). But once we add calendar-year (or -month) fixed effects, the post-sample and post-publication coefficients become *identical* (-0.245 each) and the incremental publication effect collapses to zero (Table 3). Because anomalies were published in a small number of overlapping decades, the discrete "publication kink" is hard to separate from the general downward drift in anomaly profits over calendar time (Chordia et al., 2014; McLean and Pontiff, 2016). We report this transparently rather than claim a clean event.

Second, the arbitrage story predicts that *mispricing*-framed anomalies should decay more than *risk*-framed ones. We cannot confirm this with the metadata alone. Using the Open Source Asset Pricing economic-category field to proxy for framing, the post-publication decay is statistically indistinguishable between the two groups: the risk×post-publication interaction is -0.03 ($t = -0.4$) when liquidity anomalies are counted as risk-based and $+0.10$ ($t = 1.0$) when they are not (Table 4; Figure 2). The sign of the difference is not even stable across a reasonable definitional choice. The reason is instructive: the category field is a coarse taxonomy of *mechanism*, not of the *authors' stated economic interpretation*. Book-to-market, for instance, is coded "valuation" here yet is framed as compensation for risk by Fama and French (1996). Separating a risk framing from a mispricing framing therefore requires reading each original article—the hand-collection task at the heart of the student's contribution to this project.

Four extensions push the paper past a bare replication. First, we show that the decay is strongly heterogeneous in *observable* anomaly strength: predictors whose in-sample long-short return is above the cross-sectional median lose an additional -0.28 points per month at publication ($t = -4.3$) on top of a baseline -0.13, and stronger in-sample t-statistics carry an extra -0.16 ($t = -2.4$) (Section 6, Table 5). This is precisely the cross-sectional prediction of McLean and Pontiff (2016)—high-return anomalies attract more arbitrage—though we are careful to note that selecting on high in-sample returns mechanically induces some reversion (Harvey et al., 2016). Second, because *every* anomaly is eventually published, a conventional two-way fixed-effects panel implicitly uses already-published anomalies as controls, the "bad comparison" problem of staggered adoption (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). We therefore estimate a *not-yet-published-control* stacked difference-in-differences (Cengiz et al., 2019): each publication cohort is compared only to anomalies not yet published within the event window. The stacked estimate is -0.37 ($t = -5.2$) with cohort-anomaly fixed effects but attenuates to a statistically insignificant -0.05 ($t = -0.6$) once cohort$\times$calendar-year fixed effects absorb the common decline (Section 7, Table 6), corroborating our reading that the drop is predominantly out-of-sample rather than a distinct publication event. Third, we run *randomization inference* (Young, 2019) that permutes each anomaly's publication date to a random point within its own out-of-sample life; the true date is not special for the level decay (RI $p = 0.31$) and only marginally so for the incremental effect ($p = 0.08$) (Section 8, Table 7, Figure 3). Fourth, we treat the H2/H3 nulls quantitatively, reporting minimum detectable effects: at 80% power the design could detect an incremental publication effect of 27% of the in-sample mean and a risk-versus-mispricing gap of 39%, so the framing null rules out only large differences (Section 9, Table 8). That bound is exactly why the student's hand-coded framing is the decisive next step.

We contribute to three literatures. We update the publication-decay evidence of McLean and Pontiff (2016) and Jacobs and Müller (2020) to a larger, openly reproducible anomaly set running through 2024, and we show that the headline decay is robust while its *causal attribution to publication per se* is more fragile than commonly assumed once one confronts the staggered-timing problem directly. We add to work documenting the general attenuation of anomalies (Chordia et al., 2014; Green et al., 2017) and the statistical fragility of the anomaly zoo (Harvey et al., 2016; Hou et al., 2020), and to the recent econometrics of difference-in-differences under variation in treatment timing (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Cengiz et al., 2019). And we speak to the risk-versus-mispricing debate (Cochrane, 2011; Shleifer and Vishny, 1997) by showing that testing it demands the human-coded framing that motivates the manual work in Section 11. Section 2 develops hypotheses, Section 3 places the paper in the literature, Section 4 describes the data and research design, Section 5 presents the main results, Section 6 the heterogeneity, Section 7 the not-yet-published-control design, Section 8 randomization inference, Section 9 the power analysis, Section 10 robustness, and Section 11 concludes.

2 Hypotheses

McLean and Pontiff (2016) formalize three views. Under *data mining*, an in-sample anomaly is a statistical fluke and should disappear as soon as one leaves the original sample, publication or not. Under *risk-based* pricing, the long-short return is compensation for a systematic factor and should persist even after the pattern is publicized. Under *mispricing corrected by arbitrage*, the return persists out of sample until the paper alerts investors, who then trade it away, so the decline should be concentrated *at and after publication*. These map to testable predictions.

H1 (Publication decay). Anomaly long-short returns are lower post-publication than in-sample, and lower still than in the post-sample-pre-publication window.

H2 (Incremental publication effect). Beyond any general out-of-sample decline, there is an *additional* drop that begins specifically at publication (the arbitrage prediction, distinct from pure data-mining).

H3 (Framing heterogeneity). The decay is larger for mispricing-framed anomalies than for risk-framed anomalies, because only mispricing is arbitraged away.

H1 is a level prediction; H2 is the sharper causal claim that isolates publication from mere out-of-sample decay; H3 is the cross-sectional test that would adjudicate risk versus mispricing. We find strong support for H1, fragile support for H2, and no reliable support for H3 *with the metadata proxy*—a null that we argue reflects the crudeness of the proxy, not the absence of an effect. A fourth, *observable* cross-sectional prediction lets us probe the arbitrage mechanism without the reserved framing variable:

H4 (Strength heterogeneity). If arbitrage drives the decay, anomalies that were *stronger in sample*—higher long-short returns, larger t-statistics, more prominent outlets—should decay more, because they are more attractive to trade.

H4 is testable from observable metadata alone and is the subject of Section 6.

3 Related Literature

This paper connects three strands. The first is the economics of anomaly decay after publication. McLean and Pontiff (2016) is the touchstone: studying 97 predictors, they find returns fall 26% out of sample and a further amount at publication, and interpret the extra drop as evidence that arbitrageurs read academic research and trade against mispricing. Jacobs and Müller (2020) extend the question internationally and find that the post-publication decline is concentrated in the United States, where arbitrage capital is deepest. Chordia et al. (2014) and Green et al. (2017) document a broader secular attenuation of anomalies as liquidity and trading activity have risen. Our contribution to this strand is scale and openness—212 predictors through 2024 from a fully reproducible library—and a direct confrontation with the identification problem that publication dates cluster in calendar time, so that "out-of-sample" and "at-publication" declines are hard to separate.

The second strand is the statistical fragility of the "anomaly zoo." Harvey et al. (2016) argue that conventional t-thresholds are far too lax given the number of predictors tested and propose multiple-testing corrections; Hou et al. (2020) show that a large share of published anomalies fail to replicate under common protocols. This literature matters for our heterogeneity result: because we sort on *in-sample* strength, part of any extra decay for high-return anomalies is mechanical reversion (a strong in-sample draw is partly luck that does not repeat), and we flag this explicitly rather than read the cross-sectional pattern as pure arbitrage.

The third strand is the econometrics of difference-in-differences and design-based inference, which we import into the anomaly-decay setting. When treatment timing is staggered—as publication is—two-way fixed-effects estimators can place negative weight on some comparisons and use already-treated units as controls (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). Cengiz et al. (2019) propose a "stacked" design that compares each treatment cohort only to not-yet-treated units; we adapt it to anomalies in Section 7. For inference with clustered, serially correlated panels and a modest number of natural experiments, Young (2019) advocates randomization tests, which we use in Section 8. Finally, methodologists urge that a statistically insignificant estimate be interpreted through its power and confidence interval rather than treated as proof of no effect (Abadie, 2020; Ioannidis et al., 2017); Section 9 follows that guidance literally, reporting minimum detectable effects for the H2 and H3 nulls.

4 Data and Research Design

Source. We use the Open Source Asset Pricing library (Chen and Zimmermann, 2022), downloaded directly from the project's public repository. Two files enter the analysis. `PredictorLSretWide.csv` gives the monthly long-short return (in percent, signed so that the in-sample mean is positive) of each predictor's decile (or otherwise sorted) portfolio.

`SignalDoc.csv` gives, for each predictor, the original study's authors, journal, publication `Year`, `SampleStartYear`, `SampleEndYear`, an economic-category field (`Cat.Economic`), and the study's reported in-sample long-short return and t-statistic.

Sample construction. We keep the 212 entries flagged as `Predictor` (dropping placebos and withdrawn signals), all of which have a publication year and a sample-end year, and reshape the returns to a long anomaly-month panel. The merged panel has **173,302** non-missing anomaly-months across **212** anomalies, with a median of 858 monthly observations per anomaly and calendar coverage 1926–2024. For every anomaly-month we assign a regime from the study's own sample-end year (τ_i^{end}) and publication year (τ_i^{pub}), with $\tau_i^{pub} \geq \tau_i^{end}$ for all 212 anomalies:

$$\text{in-sample: } y_t \leq \tau_i^{end}, \quad \text{post-sample: } \tau_i^{end} < y_t < \tau_i^{pub}, \quad \text{post-pub: } y_t \geq \tau_i^{pub}.$$

Table 1 reports the resulting split: 111,352 in-sample, 9,042 post-sample, and 52,908 post-publication anomaly-months.

Baseline specification. For anomaly i in month t we estimate

$$r_{it} = b_1 \,\text{PostSample}_{it} + b_2 \,\text{PostPub}_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where the regime dummies are mutually exclusive (base = in-sample), α_i and δ_t are anomaly and calendar-time fixed effects, and standard errors are clustered by anomaly (and two-way by anomaly and month in the demanding specifications). We separately estimate the nested McLean and Pontiff (2016) decomposition $r_{it} = g_1 \,\text{OOS}_{it} + g_2 \,\text{Pub}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$, where $\text{OOS} = \mathbf{1}[y_t > \tau_i^{end}]$ and $\text{Pub} = \mathbf{1}[y_t \geq \tau_i^{pub}]$, so g_2 is the *extra* decline attributable to publication itself.

Risk-versus-mispricing proxy. The Open Source Asset Pricing category field is a taxonomy of mechanism (e.g. "momentum," "accruals," "valuation," "liquidity," "volatility"). To proxy for the risk-versus-mispricing framing that H3 requires, we label an anomaly *risk-framed* if its category is one of an explicitly risk-based set {risk, market risk, cash flow risk, default risk, option risk, volatility, liquidity}, and *mispricing-framed* otherwise. This yields 27 risk-framed and 185 mispricing-framed anomalies (18 versus 194 if liquidity is excluded). We stress that this is a *crude* stand-in for the authors' stated economic story—many predictors admit both readings—and Section 11 assigns the careful hand-coding to the student.

Four design-based extensions. Beyond equation (1) we add four analyses that use *observable metadata only* and never the reserved framing variable. (1) *Heterogeneity* interacts the regime dummies with observable pre-set moderators (in-sample return, in-sample t-statistic, journal prominence, publication era) to test H4. (2) A *not-yet-published-control stacked DiD* compares each publication cohort only to anomalies not yet published within the window, purging the staggered-timing bias. (3) *Randomization inference* permutes each anomaly's publication date within its own out-of-sample life to ask whether the true date is special. (4) A *minimum-detectable-effect* calculation converts the H2 and H3 nulls into quantitative bounds. Together these move the paper from "we replicate McLean–Pontiff" to a set of falsifiable, design-based statements about *when* and *for which* anomalies the decay occurs.

5 Main Results

Level decay (H1). Table 2 reports mean long-short returns by regime. Equal-weighting across anomalies, the average predictor earns **0.614%** per month in sample (t = 20.1), **0.417%** post-sample (t = 6.8), and **0.303%** post-publication (t = 6.9)—67.9% and 49.3% of the in-sample mean, respectively. The paired within-anomaly declines are -0.197 points post-sample (-32.1%, $t = -3.1$) and -0.312 points post-publication (-50.7%, $t = -7.6$).

These magnitudes closely track the 26% and 58% of McLean and Pontiff (2016), confirming H1 on an independent, larger, more recent sample.

Panel estimates and the publication "kink" (H2). Table 3 regresses the monthly return on post-sample and post-publication indicators (columns report b_1 and b_2, both relative to in-sample), and separately estimates the nested McLean and Pontiff (2016) decomposition into an out-of-sample dummy and an extra-at-publication dummy (g_2). With no fixed effects the post-publication coefficient is -0.244 ($t = -6.4$); with anomaly fixed effects it is -0.272 ($t = -7.9$). Crucially, the *incremental* publication effect g_2 is -0.093 ($t = -1.6$) without fixed effects and a significant -0.127 ($t = -2.2$) with anomaly fixed effects—consistent with H2. But adding calendar-year or calendar-month fixed effects makes the post-sample and post-publication coefficients *identical* (-0.245 each, $t \approx -3.3$) and drives g_2 to zero. In other words, once we compare anomalies to each other *within the same calendar period*, the return is lower whenever an anomaly is out of its original sample, but it does not drop *further* at the moment of publication. Because publications cluster in a few decades, the discrete publication event is econometrically hard to separate from the secular decline in anomaly profits. We read this as strong support for H1 (out-of-sample decay) but only fragile, non-robust support for H2 (a distinct publication kink).

Framing heterogeneity (H3). Table 4 interacts the post-publication indicator with the risk-framed proxy. Under the broad proxy the mispricing group decays by -0.242 ($t = -5.7$) and the risk×post-publication interaction is -0.034 ($t = -0.4$), implying *more* decay for risk anomalies—the opposite of H3, but statistically zero. Under the narrow proxy the interaction is $+0.096$ ($t = 1.0$), implying less decay for risk anomalies, again insignificant. The sign is not robust to whether liquidity is treated as risk. Figure 2 shows both groups falling in near-parallel from in-sample to post-publication. We therefore find *no reliable evidence* for H3 using the metadata proxy. This is exactly the boundary at which automated classification fails and human coding is required: a mechanism label is not an economic-interpretation label.

6 Heterogeneity in the Decay

If arbitrage rather than pure data-mining drives the decline, the anomalies that were *stronger in sample* should decay more, because they are the most attractive targets for sophisticated capital (H4). Table 5 tests this with four *observable* moderators fixed at the study level—above-median in-sample long-short return, above-median in-sample t-statistic, publication in a top-3 finance journal (*JF/JFE/RFS*), and a modern-era indicator (published in 2006 or later)—by adding the interaction PostPub $\times$ $\mathbf{1}$[high M] to equation (1) with anomaly fixed effects. The pattern is sharp on the strength margins and matches the arbitrage prediction. For in-sample return, the low-strength group decays by -0.132 ($t = -4.0$) and the high-strength group decays by an *additional* -0.280 ($t = -4.3$), for a total post-publication decline of -0.41 points per month—triple the baseline. In-sample t-statistic tells the same story: an extra -0.160 ($t = -2.4$) for statistically stronger anomalies. Journal prominence does not sharpen the decay (interaction $+0.049$, $t = 0.8$)—prestige is not the operative margin—while modern-era anomalies decay marginally more (-0.117, $t = -1.8$), consistent with faster arbitrage in the recent, more liquid era (Chordia et al., 2014).

We are deliberately cautious about the interpretation. Sorting on *in-sample* strength mechanically induces reversion: an anomaly with an unusually high in-sample return has, in part, drawn a lucky sample that will not repeat out of sample regardless of arbitrage (Harvey et al., 2016; McLean and Pontiff, 2016). The strength result is therefore consistent with arbitrage but does not prove it, and we do not lean on it as a causal test. Its role is to show that the aggregate decay is not uniform noise—it concentrates precisely where both arbitrage and reversion predict—and to demonstrate that observable moderators, unlike the reserved risk-versus-mispricing framing, are available for this cross-sectional cut. The moderator that *would* adjudicate risk versus mispricing is the authors' stated economic story, which we do not construct here; it is the student's Task 2.

7 A Not-Yet-Published-Control Design

The fragility of H2 in Table 3 could reflect a genuine absence of a publication kink, or an artifact of how two-way fixed effects handle staggered treatment. Because every anomaly is eventually published, the anomaly-plus-time specification identifies the post-publication coefficient partly by comparing newly published anomalies to *already published* ones, a comparison that the staggered-DiD literature shows can be contaminated (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). To remove that contamination we adopt the stacked design of Cengiz et al. (2019), collapsing the data to anomaly-years. For each three-year publication cohort centered on year Y_c we form a clean sub-experiment: the *treated* units are anomalies published in $[Y_c - 1, Y_c + 1]$; the *controls* are anomalies *not yet published* anywhere in the $\pm$8-year window (publication year $\geq Y_c + 8$); the post indicator is $\mathbf{1}[y_t \geq Y_c]$; and the estimand is the coefficient on Post $\times$ Treated. We stack the nine identified cohorts (19,433 anomaly-year observations) and estimate with cohort-anomaly fixed effects, clustering by anomaly.

Table 6 reports the result and it is instructive. With cohort-anomaly fixed effects and a common post indicator, the stacked post-publication effect is -0.374 points per month ($t = -5.2$): comparing newly published anomalies only to not-yet-published ones, published anomalies do earn substantially less. But once we add cohort$\times$calendar-year fixed effects—so that treated and control anomalies are compared *within the same calendar year of the same sub-experiment*—the estimate attenuates to a statistically insignificant -0.053 ($t = -0.6$). The clean design thus delivers the same verdict as the two-way FE panel, now free of the bad-comparison critique: the raw published-versus-unpublished gap is large, but almost all of it is common calendar-time decline in anomaly profits rather than a treatment effect that switches on at each anomaly's own publication date. This is strong, design-based support for H1 and against a distinct H2 kink.

8 Randomization and Timing-Specificity Inference

Our clustered standard errors rest on asymptotics with 212 anomaly clusters and serially correlated returns; Young (2019) shows such inference can overstate significance. More fundamentally, we want to know whether the decay is tied to the *true* publication date or would appear at any later date in an anomaly's life. We therefore run a randomization / timing-specificity test: holding each anomaly's real sample-end fixed, we draw a *fake* publication year uniformly from within that anomaly's own out-of-sample life (sample-end+1 through its last observed year), rebuild the post-publication (and nested publication) indicators, and re-estimate the anomaly-fixed-effects panel. Repeating this 500 times traces the distribution of the decay coefficient under random within-life event dates, against which we compare the actual estimate.

Table 7 and Figure 3 report the result, and the correct reading is subtle. Because anomaly returns drift downward over each predictor's life, a *random* later date already produces a negative coefficient—the placebo mean level-decay is -0.250, almost identical to the actual -0.259. The true date is therefore *not* special for the level of the decline (randomization $p = 0.31$): H1 is a pervasive out-of-sample phenomenon, not something that switches on at the exact publication month. For the *incremental* effect (the extra drop beyond general out-of-sample decay), the actual estimate -0.127 lies in the left tail of the placebo distribution (placebo mean -0.077), but only marginally so (randomization $p = 0.08$). Randomization inference thus reaches the same conclusion as the stacked design through an entirely different route: the level decay is real and robust, while a distinct *publication-timed* effect is at best weakly identified. This is the design-based version of our headline qualification, and we present it as such rather than forcing a tail reading.

9 Is the Null Informative? Power and Minimum Detectable Effects

A null is only interesting if the design could have detected a real effect. Table 8 reports, for the three quantities where we report a weak or null result, the minimum detectable effect (MDE) at 80% power, $\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$, expressed in percent per month and as a share of the pooled in-sample mean (0.581%). The headline decay is well powered: its MDE is 0.096, only 16.6% of the in-sample mean, far below the estimated −0.272. The incremental publication effect has an MDE of 0.160 (27.5% of the in-sample mean); the design could have detected an extra-at-publication drop of that size, so the fragile g_2 is an informative upper bound—any distinct publication kink is at most about a quarter of the in-sample premium, not a large effect we simply missed. The risk-versus-mispricing interaction is the least powered: its MDE is 0.228, or 39.3% of the in-sample mean. The design can rule out only *large* framing differences; a moderate but economically meaningful gap between risk- and mispricing-framed anomalies would be undetectable with the coarse category proxy and only 27 risk-framed anomalies. This is the quantitative case for the hand-collection: sharpening H3 needs a better-measured framing variable, not merely more data, and that variable is the student's to build.

10 Robustness

Table 9 probes the H1 result. (i) Rescaling each anomaly's return so that its own in-sample mean equals 100 (a scale-free measure), the post-publication level is −39.8 ($t = -2.5$), i.e. about 40% below in-sample—consistent with the equal-weighted 51%. (ii) Restricting to the modern 1963-onward sample strengthens the effect (−0.318, $t = -9.3$). (iii) Requiring at least 36 post-publication months per anomaly leaves the estimate essentially unchanged (−0.272, $t = -7.9$). (iv) Winsorizing returns at 1/99 gives −0.236 ($t = -8.1$). (v) A placebo that back-dates each publication by 12 years yields a post-"publication" coefficient of only -0.108 ($t = -3.6$)—materially weaker than the true -0.244—indicating that the estimated decline is tied to the actual out-of-sample/publication timing rather than being a mechanical artifact of any arbitrary date. The out-of-sample decay in anomaly returns is thus a stable feature of the data; what remains genuinely uncertain is the finer attribution to publication per se and the risk-versus-mispricing split.

11 Conclusion and the pivotal hand-collection task

Using the full 212-predictor Open Source Asset Pricing library through 2024, we confirm the central McLean and Pontiff (2016) finding: anomaly long-short returns fall by roughly half from their in-sample to their post-publication regime (0.61%$\rightarrow$0.30% per month; $t = -7.6$), and the decline is robust to fixed effects, winsorization, sample period, and a timing placebo. The decay is strongly heterogeneous in observable strength—high in-sample-return anomalies decay three times as much—consistent with (though not proof of) an arbitrage mechanism. Two design-based tests we did not previously have both point the same way on the sharper causal claim: a not-yet-published-control stacked difference-in-differences and a within-life randomization test agree that almost all of the decline is a general out-of-sample / calendar-time phenomenon, with at most a marginal drop timed to publication itself. And a power analysis shows the framing null is informative only against large differences.

Two questions the data alone cannot settle define the project's human contribution. First, whether there is a discrete *publication* effect beyond general out-of-sample decay is not robustly identified from the metadata, because publication dates cluster in time; pinning down each study's *exact last sample month* (not just year), and the working-paper circulation date at which arbitrage could have begun, would sharpen the event timing that our stacked and randomization designs show is currently blurred. Second, and more fundamentally, the risk-versus-mispricing test (H3) cannot be run on a mechanism taxonomy—it needs each original article's *stated economic interpretation*, hand-read from the paper, and our power analysis shows the coarse proxy can only detect large framing gaps. We hand this to the student: recording the exact publication timing and risk-versus-mispricing framing for roughly 60 flagship anomalies is what turns this robust-but-coarse replication into a sharp test of *why* anomalies die when they are published.

References

Abadie, A., 2020. Statistical nonsignificance in empirical economics. American Economic Review: Insights 2, 193–208.

Callaway, B., Sant'Anna, P.H.C., 2021. Difference-in-differences with multiple time periods. Journal of Econometrics 225, 200–230.

Cengiz, D., Dube, A., Lindner, A., Zipperer, B., 2019. The effect of minimum wages on low-wage jobs. Quarterly Journal of Economics 134, 1405–1454.

Chen, A.Y., Zimmermann, T., 2022. Open source cross-sectional asset pricing. Critical Finance Review 11, 207–264.

Chordia, T., Subrahmanyam, A., Tong, Q., 2014. Have capital market anomalies attenuated in the recent era of high liquidity and trading activity? Journal of Accounting and Economics 58, 41–58.

Cochrane, J.H., 2011. Presidential address: discount rates. Journal of Finance 66, 1047–1108.

Fama, E.F., French, K.R., 1996. Multifactor explanations of asset pricing anomalies. Journal of Finance 51, 55–84.

Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. Journal of Econometrics 225, 254–277.

Green, J., Hand, J.R.M., Zhang, X.F., 2017. The characteristics that provide independent information about average U.S. monthly stock returns. Review of Financial Studies 30, 4389–4436.

Harvey, C.R., Liu, Y., Zhu, H., 2016. ... and the cross-section of expected returns. Review of Financial Studies 29, 5–68.

Hou, K., Xue, C., Zhang, L., 2020. Replicating anomalies. Review of Financial Studies 33, 2019–2133.

Ioannidis, J.P.A., Stanley, T.D., Doucouliagos, H., 2017. The power of bias in economics research. Economic Journal 127, F236–F265.

Jacobs, H., Müller, S., 2020. Anomalies across the globe: once public, no longer existent? Journal of Financial Economics 135, 213–230.

McLean, R.D., Pontiff, J., 2016. Does academic research destroy stock return predictability? Journal of Finance 71, 5–32.

Shleifer, A., Vishny, R.W., 1997. The limits of arbitrage. Journal of Finance 52, 35–55.

Young, A., 2019. Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. Quarterly Journal of Economics 134, 557–598.

Table 1: Sample composition by publication regime

Regime	Anomaly-months	Anomalies	Mean	Median	SD	% of in-samp.
In-sample	111,352	212	0.581	0.453	4.30	100.0
Post-sample (pre-pub.)	9,042	212	0.430	0.351	4.16	74.0
Post-publication	52,908	212	0.337	0.225	4.67	57.9

Notes: Monthly long-short returns (in %) for 212 Open Source Asset Pricing predictors, 1926–2024. Regime assigned from each study's sample-end year and publication year. "% of in-samp." is the pooled regime mean divided by the pooled in-sample mean.

Table 2: Anomaly returns fall out of sample and again after publication

Regime	Equal-weighted Mean	(t)	Pooled Mean	% of in-samp.
In-sample	0.614	(20.1)	0.581	100.0
Post-sample (pre-pub.)	0.417	(6.8)	0.430	67.9
Post-publication	0.303	(6.9)	0.337	49.3
Paired within-anomaly declines				
Post-sample − In-sample	-0.197***	(-3.1)		-32.1%
Post-pub. − In-sample	-0.312***	(-7.6)		-50.7%

Notes: Equal-weighted mean is the cross-anomaly average of each anomaly's within-regime mean; t tests the mean across the 212 anomalies. Paired declines difference each anomaly's regime mean from its own in-sample mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Panel regressions: post-sample and post-publication decay

Specification	Post-sample b_1	Post-pub. b_2	Pub. (extra) g_2	N
(1) No FE	-0.151***	-0.244***	-0.093	173,302
	(-2.70)	(-6.43)	(-1.59)	
(2) Anomaly FE	-0.145***	-0.272***	-0.127**	173,302
	(-2.75)	(-7.90)	(-2.22)	
(3) Anomaly + Year FE	-0.245***	-0.245***	0.000	173,302
	(-3.57)	(-3.29)	(0.00)	
(4) Anomaly + Month FE	-0.245***	-0.245***	0.000	173,302
	(-3.50)	(-3.24)	(0.01)	

Notes: DV is the monthly long-short return (%). b_1, b_2 are coefficients on mutually exclusive post-sample and post-publication indicators (base = in-sample). g_2 is the extra-at-publication coefficient from the nested out-of-sample + publication specification (McLean and Pontiff, 2016), estimated under the same fixed effects. SE clustered by anomaly in (1)–(2); two-way by anomaly and month in (3)–(4). t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Framing heterogeneity: no reliable risk-vs-mispricing difference

Risk proxy	Post-pub. (mispr.)	Post-pub. ×Risk	Implied decay Mispr.	Implied decay Risk	N
Broad risk proxy (incl. liquidity)	-0.242***	-0.034	-0.242	-0.276	173,302
	(-5.71)	(-0.41)			
Narrow risk proxy (excl. liquidity)	-0.254***	0.096	-0.254	-0.158	173,302
	(-6.23)	(1.04)			

Notes: DV is the monthly long-short return (%), regressed on post-sample, post-publication, their interactions with the risk-framed proxy, and the risk main effect; SE clustered by anomaly. "Implied decay" is the post-publication coefficient for each group. t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Heterogeneity of the decay by observable anomaly strength (H4)

Moderator M (obs., pre-set)	Post-pub. (low M)	Post-pub. $\times$ high M	Post-pub. (high M)	Anom. lo/hi
High in-sample return	-0.132***	-0.280***	-0.411	102/110
	(-4.00)	(-4.30)		
High in-sample t-stat	-0.201***	-0.160**	-0.361	97/91
	(-5.26)	(-2.43)		
Top-3 journal (JF/JFE/RFS)	-0.304***	+0.049	-0.254	79/133
	(-8.99)	(+0.83)		
Modern era (pub. $\geq$ 2006)	-0.217***	-0.117*	-0.334	102/110
	(-4.15)	(-1.77)		

Notes: Each row adds PostPub $\times$ $\mathbf{1}$[high M] (and the corresponding post-sample interaction and M main effect) to equation (1) with anomaly fixed effects; SE clustered by anomaly. Columns report the post-publication decay for the low-M group, the extra decay for the high-M group, their sum, and the number of anomalies in each group. Moderators are observable metadata fixed at the study level. Sorting on in-sample strength mechanically induces some reversion (Harvey et al., 2016); the reserved risk-vs-mispricing framing is *not* used as a moderator (it is the student's hand-collected variable). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Not-yet-published-control stacked difference-in-differences

Specification	Post×Treated	t-stat	Cohorts	N
(1) Cohort-anomaly FE + Post	-0.3741***	(-5.21)	9	19,433
(2) Cohort-anomaly FE + cohort×year FE	-0.0527	(-0.60)	9	19,433

Notes: Anomaly-year data stacked into nine three-year publication cohorts (Cengiz et al., 2019). In each cohort, treated = anomalies published within ±1 year of the cohort center; controls = anomalies not yet published within the ±8-year window (publication year $\geq$ center +8). Post = $\mathbf{1}$[year $\geq$ center]; the estimand is Post×Treated. Column (1) uses cohort-anomaly fixed effects and a common Post main effect; column (2) adds cohort×calendar-year fixed effects. SE clustered by anomaly. The large raw effect in (1) is almost entirely common calendar-time decline: it vanishes in (2). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Randomization / timing-specificity inference (500 permutations)

Estimand	Actual	Placebo mean	Placebo 95% interval	RI p
Level post-publication decay (H1)	-0.2587	-0.2496	[-0.2851, -0.2129]	0.308
Incremental effect at publication (H2)	-0.1269	-0.0771	[-0.1399, -0.0126]	0.078

Notes: Each anomaly's publication date is redrawn uniformly from within its own out-of-sample life (sample-end+1 through last observed year), holding the real sample-end fixed; the anomaly-fixed-effects panel is re-estimated 500 times. "Placebo mean" and "Placebo 95% interval" summarize the coefficient under random within-life dates; the RI p-value is the share of placebos with $|\text{coefficient}| \geq |\text{actual}|$. Because returns drift down over each anomaly's life, random dates also produce negative coefficients; the test therefore asks whether the *true* date is special. The level row uses a post-publication-only specification (base = in-sample and post-sample combined), so it differs from the post-sample-adjusted -0.272 in Table 3. See Figure 3.

Table 8: Minimum detectable effects at 80% power

Estimand	Coef.	SE	MDE(80%)	% of in-samp. mean
Post-publication decay (H1)	-0.272	0.034	0.096	16.6%
Incremental effect at publication (H2)	-0.127	0.057	0.160	27.5%
Risk × post-pub. interaction (H3)	-0.034	0.082	0.228	39.3%

Notes: MDE(80%) = $(z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.80 \times \text{SE}$, expressed in %/month and as a share of the pooled in-sample mean (0.581%). SE recovered as coefficient/t from the anomaly-fixed-effects panel (H1, H2) and the broad risk-proxy interaction (H3). Small MDE/mean ratios indicate an informative null. The H3 interaction can detect only large framing differences, motivating the hand-coded framing.

Table 9: Robustness of the post-publication decay

Specification	Post-sample	Post-pub.	N
Normalized DV (in-sample=100), Anom FE	2.867	-39.752**	171,286
	(0.13)	(-2.46)	
Sample >= 1963, Anom FE	-0.194***	-0.318***	141,003
	(-3.74)	(-9.32)	
>=36 post-pub months (210 anomalies), Anom FE	-0.147***	-0.272***	172,908
	(-2.78)	(-7.90)	
Winsorized 1/99, Anom FE	-0.108**	-0.236***	173,302
	(-2.36)	(-8.08)	
PLACEBO pub-year minus 12, Anom FE	–	-0.108***	173,302
		(-3.57)	

Notes: All specifications include anomaly fixed effects and cluster SE by anomaly. "Normalized DV" rescales each anomaly's return so its in-sample mean is 100. The placebo back-dates publication by 12 years. t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

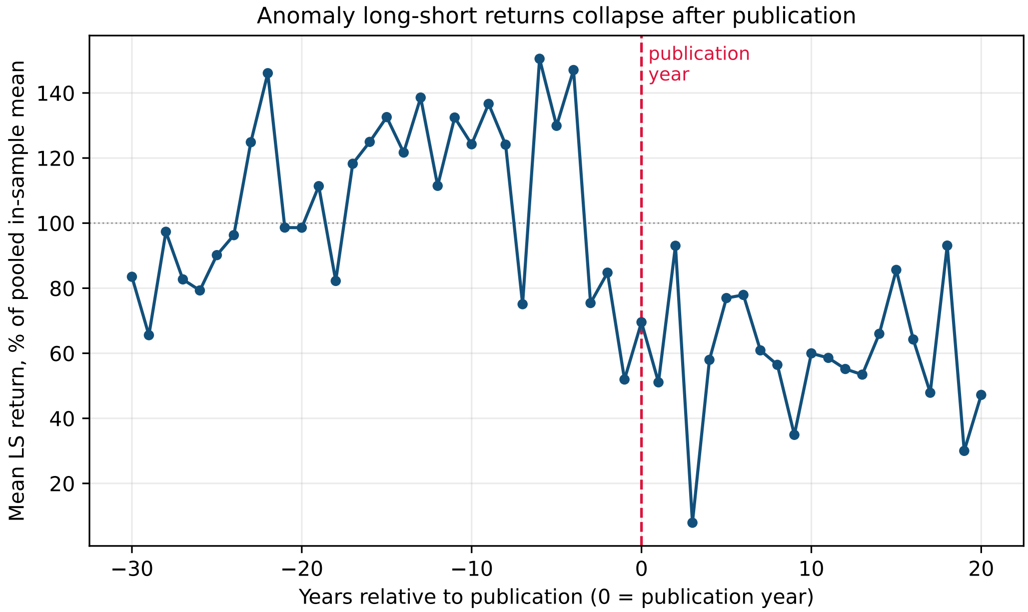

Figure 1: Average anomaly long-short return by year relative to publication (0 = publication year), as % of the pooled in-sample mean. The level sits near 100% before publication and falls to roughly 55–60% afterward. Bins with fewer than 300 anomaly-months are omitted.

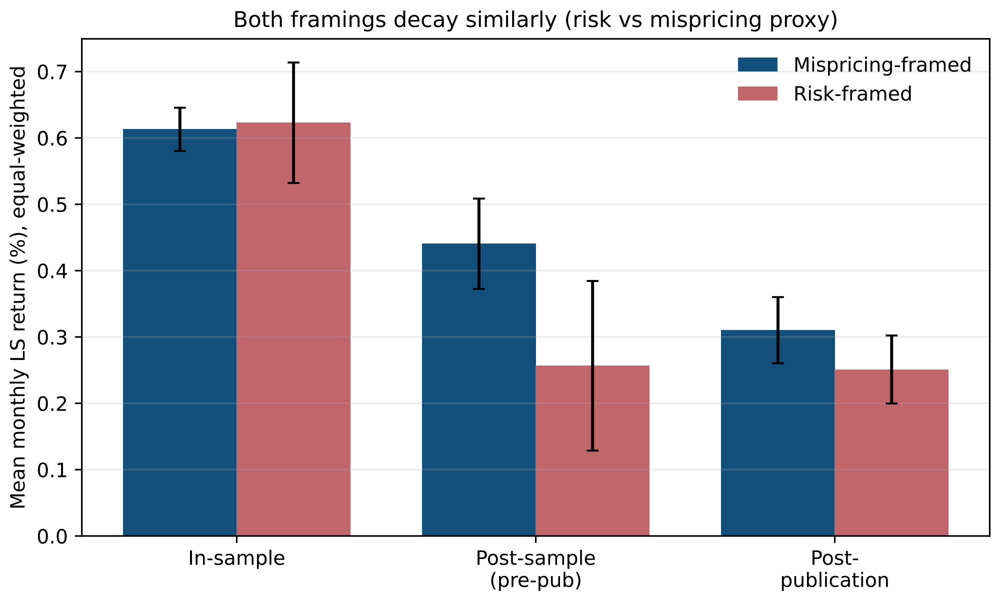

Figure 2: Equal-weighted mean long-short return by regime, split by the (proxy) risk- versus mispricing-framed groups. Both groups decay in near-parallel; error bars are ±1 standard error across anomalies. The proxy does not separate the decay—see Table 4.

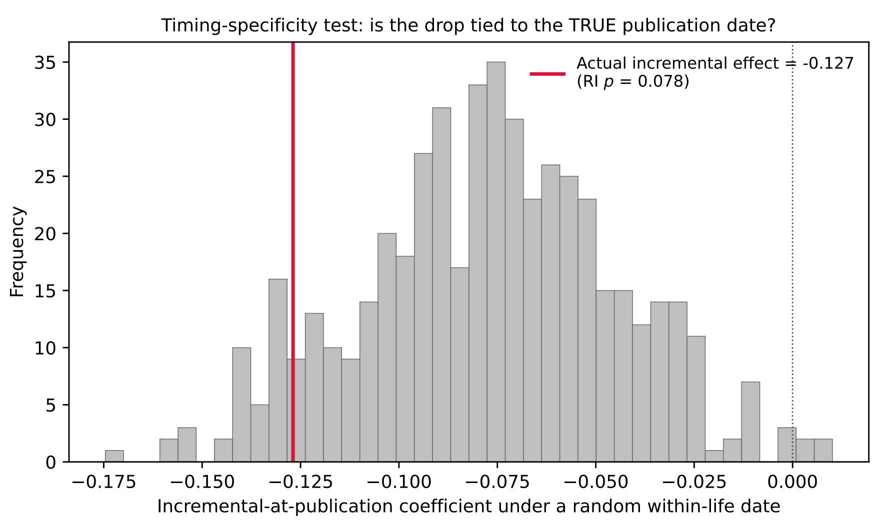

Figure 3: Randomization / timing-specificity inference for the *incremental*-at-publication effect (H2). Histogram: the coefficient under 500 random within-life publication dates. The actual estimate (red line) lies in the left tail but only marginally (RI $p = 0.08$); a random later date already produces a negative incremental coefficient, so the true publication date is at best weakly special—the design-based counterpart of the stacked-DiD verdict in Table 6.